

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO2: *Neural Network Representation, Perceptron Model, Multilayer Perceptron's, Backpropagation Algorithm, Deep Neural Networks, Genetic Algorithms, Hypothesis Space Search, Genetic Programming, Models of Evaluation and Learning (BL1, BL2)*

Assessment Tool Weightage: 10%

QUESTIONS:15

Total Marks:25

Annexure A

DEBUGGING & OPTIMIZATION TASK QUESTIONS

Code of Conduct:

I S.Rohini / 192424413 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *S.Rohini*

Annexure A

Short Answer Test

1. A genetic algorithm designed for hypothesis space search is producing inconsistent and suboptimal solutions across multiple runs. Investigate possible issues such as improper encoding of chromosomes, incorrect fitness function design, premature convergence, or lack of diversity in the population. Explain how to debug these problems systematically and suggest optimization strategies like mutation tuning, selection pressure adjustment, and elitism to enhance solution quality and convergence speed. (5 Marks)

A Genetic Algorithm (GA) may produce inconsistent results due to improper chromosome encoding, incorrect fitness function, premature convergence, low mutation rate, or lack of population diversity.

Debugging Steps:

- Verify chromosome encoding.
- Check whether the fitness function correctly evaluates solutions.
- Monitor fitness values for each generation.
- Analyze crossover and mutation operations.
- Check population diversity during execution.

Optimization Techniques:

- Increase mutation rate to improve diversity.
- Use tournament or roulette-wheel selection.
- Apply elitism to preserve the best solutions.
- Increase population size.
- Tune crossover and mutation probabilities.

Conclusion:

Proper encoding, accurate fitness evaluation, balanced mutation, and elitism improve solution quality, consistency, and convergence speed.

2. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy. (5 Marks)

A CNN may produce unstable results because of improper data augmentation, vanishing gradients, incorrect batch normalization, high learning rate, or poor weight initialization.

Debugging Steps:

- Verify image preprocessing.
- Monitor training and validation loss.
- Check gradient values.
- Validate batch normalization layers.
- Test the model on a small dataset.

Optimization Techniques:

- Use Batch Normalization.
- Apply He/Xavier initialization.
- Reduce the learning rate.
- Use the Adam optimizer.
- Apply dropout and proper data augmentation.

Conclusion:

Correct preprocessing, suitable learning rate, and proper normalization stabilize CNN training and improve image classification accuracy.

3. A support vector machine (SVM) model underperforms on a nonlinear dataset. Identify issues such as incorrect kernel choice, improper parameter tuning, or scaling problems. Suggest debugging techniques and optimization methods including kernel selection, hyperparameter tuning, and feature scaling. (5 Marks)

An SVM performs poorly on nonlinear data due to incorrect kernel selection, improper tuning of C and Gamma, lack of feature scaling, or noisy features.

Debugging Steps:

- Scale all features.
- Compare different kernels.
- Evaluate training and testing accuracy.
- Perform cross-validation.
- Analyze the confusion matrix.

Optimization Techniques:

- Use the RBF kernel.
- Normalize or standardize features.
- Tune C and Gamma using Grid Search.

- Remove irrelevant features.
- Increase the quality of training data.

Conclusion:

Selecting the correct kernel and tuning hyperparameters significantly improves SVM performance on nonlinear datasets.

4. A K-Means clustering algorithm is applied to a dataset, but the clusters formed are highly imbalanced and do not reflect the actual data distribution. Investigate possible issues such as incorrect choice of K, poor initialization of centroids, or presence of outliers. Explain debugging approaches and suggest improvements like the Elbow method, K-Means++, and outlier removal techniques. (5 Marks)

K-Means may produce poor clusters because of an incorrect value of K, poor centroid initialization, outliers, or unnormalized data.

Debugging Steps:

- Visualize clusters.
- Check centroid positions.
- Detect outliers.
- Verify feature scaling.
- Evaluate cluster quality.

Optimization Techniques:

- Use the Elbow Method to find K.
- Apply K-Means++ initialization.
- Normalize data.
- Remove outliers.
- Validate clusters using the Silhouette Score.

Conclusion:

Proper centroid initialization, correct K selection, and normalization produce balanced and accurate clusters.

5. A reinforcement learning agent trained to solve a navigation problem fails to learn an optimal policy and shows erratic behavior. Analyze possible causes such as sparse rewards, improper exploration-exploitation balance, or incorrect reward function design. Describe debugging approaches and recommend optimization strategies like reward shaping, epsilon decay tuning, and policy optimization methods to improve learning performance. (5 Marks)

A reinforcement learning agent may fail because of sparse rewards, poor exploration-exploitation balance, incorrect reward function, or improper hyperparameter settings.

Debugging Steps:

- Monitor reward values.
- Observe the agent's actions.
- Verify state transitions.
- Analyze learning curves.
- Test the reward function.

Optimization Techniques:

- Apply reward shaping.
- Tune epsilon decay.
- Optimize the learning rate and discount factor.
- Increase training episodes.
- Use policy optimization methods.

Conclusion:

Improving reward design, exploration strategy, and hyperparameter tuning enables the agent to learn an optimal policy efficiently.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	30	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	10	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand